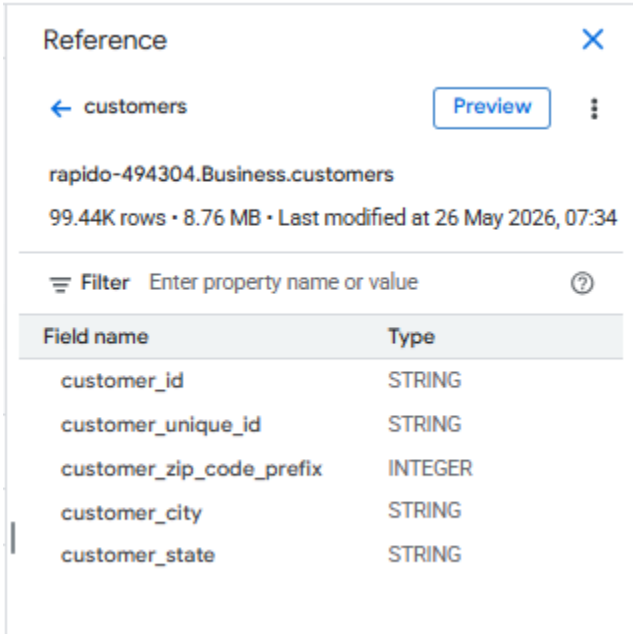
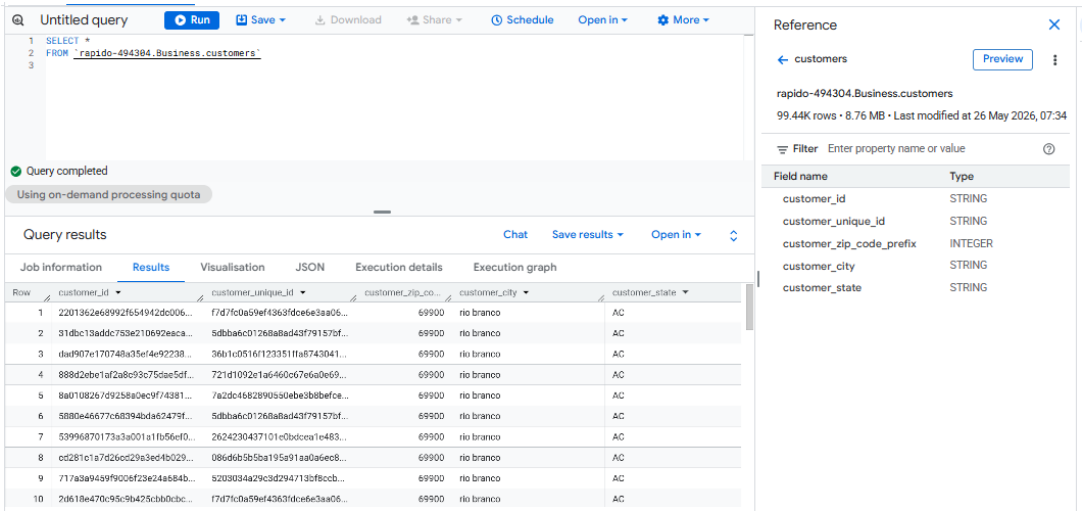
Import the dataset and do usual exploratory analysis steps like checking the structure & characteristics of the dataset:

1. **1. Data type of all columns in the "customers" table.**

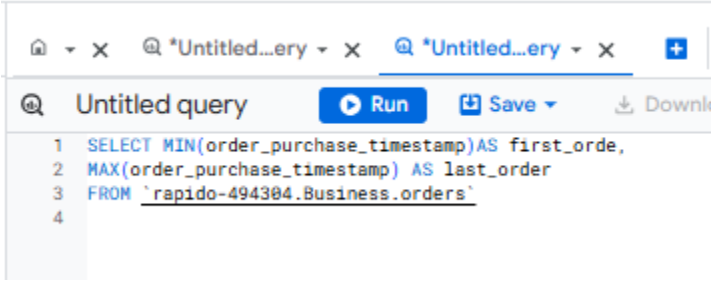




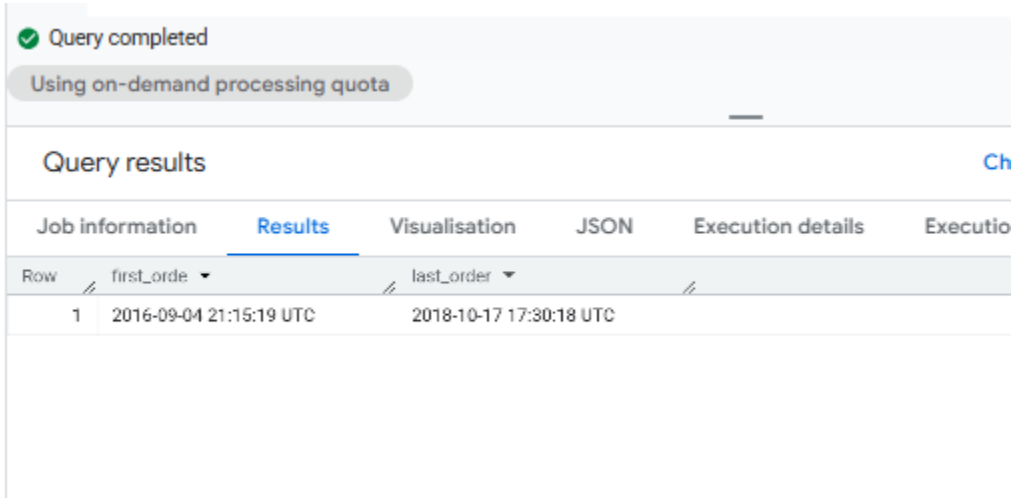
Business responses:-

2. **Get the time range between which the orders were placed.**

SQL query: -



SQL responses:-

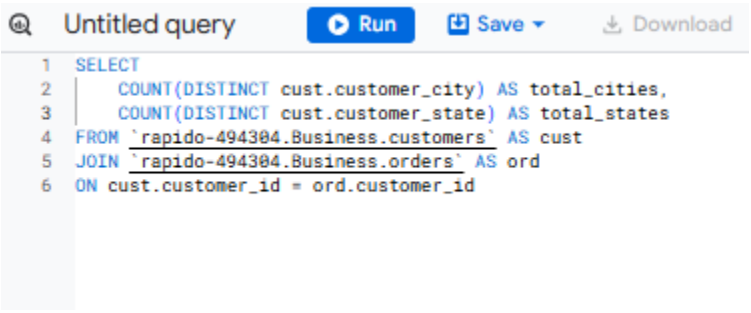


Business responses:-

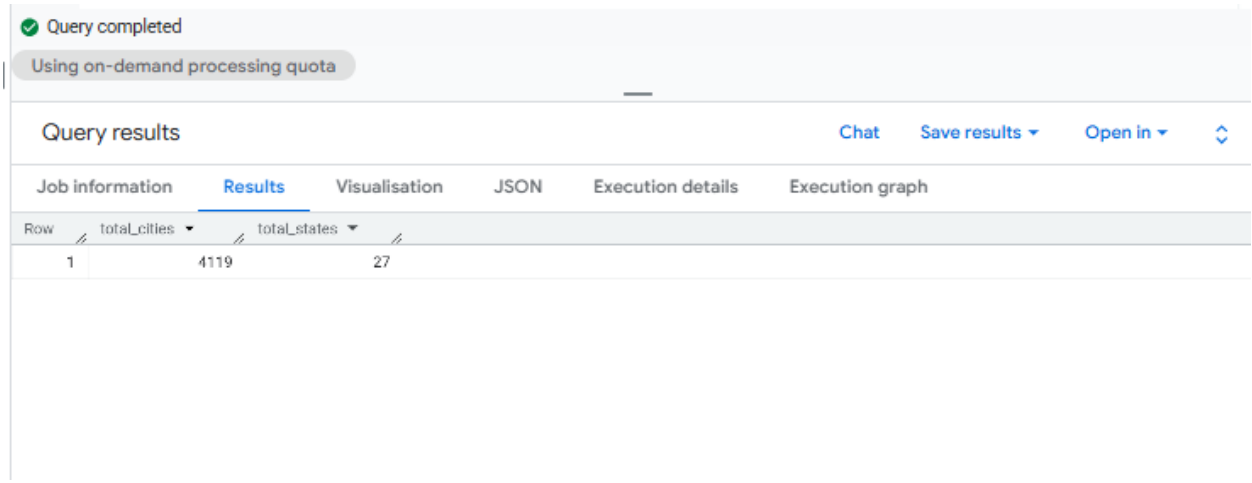
- The order data spans from September 2016 to October 2018, providing nearly two years of transaction history. This duration is adequate for identifying growth trends, seasonal patterns, and changes in customer purchasing behavior over time.

3. **Count the Cities & States of customers who ordered during the given period.**

SQL query: -



SQL responses:-



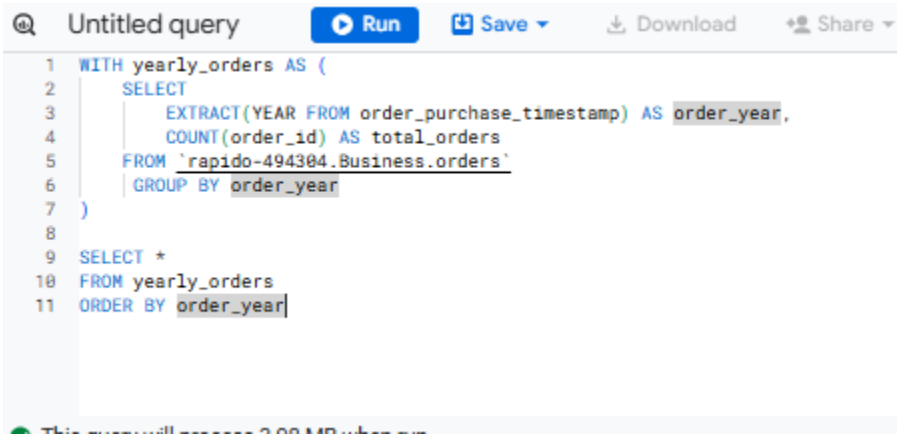
Business responses:-

- The total number of orders from 4,119 unique cities across all 27 Brazilian states shows a strong nationwide presence. This indicates successful market penetration and a diverse customer base across Brazil.

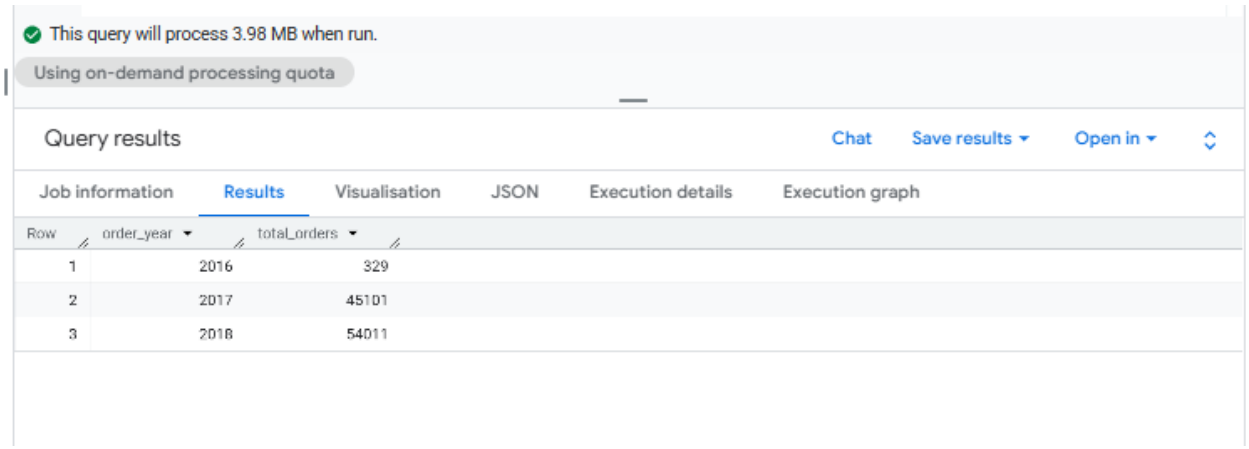
In-depth Exploration :-

1. **Is there a growing trend in the no. of orders placed over the past years?**

SQL query: -



SQL responses:-



Business responses:-

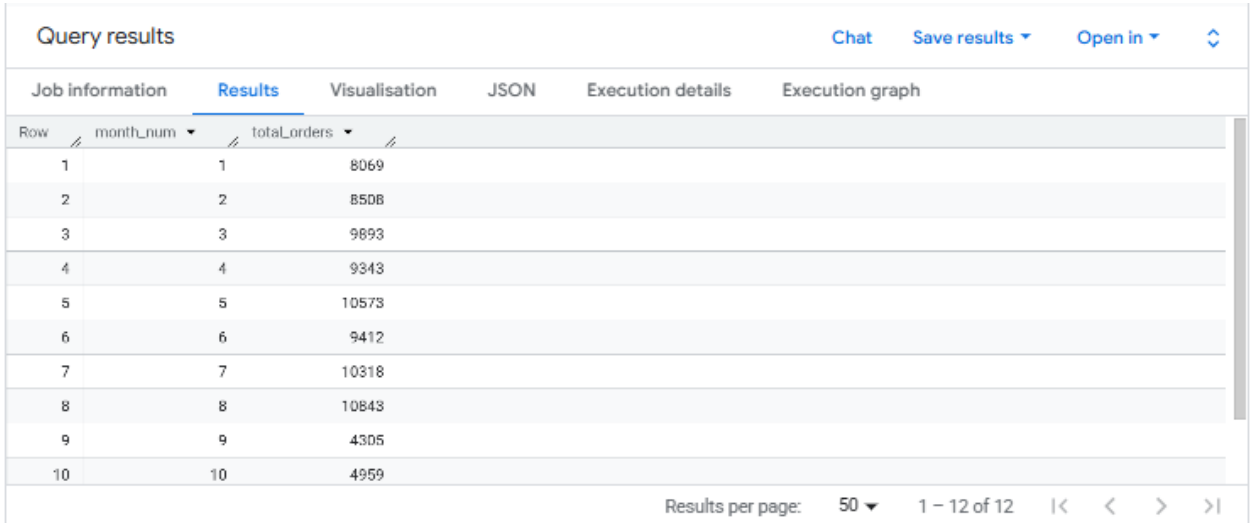
- The number of orders increased significantly year over year. While 2016 had a relatively low order volume, the business witnessed rapid expansion in 2017. Growth continued in 2018 with nearly a 20% increase in orders compared to 2017.

2.Can we see some kind of monthly seasonality in terms of the no. of orders being placed?

SQL query: -

```
SELECT
    EXTRACT(MONTH FROM order_purchase_timestamp) AS month_num,
    COUNT(order_id) AS total_orders
FROM `rapido-494304.Business.orders`
GROUP BY month_num
ORDER BY month_num;
```

SQL responses:-



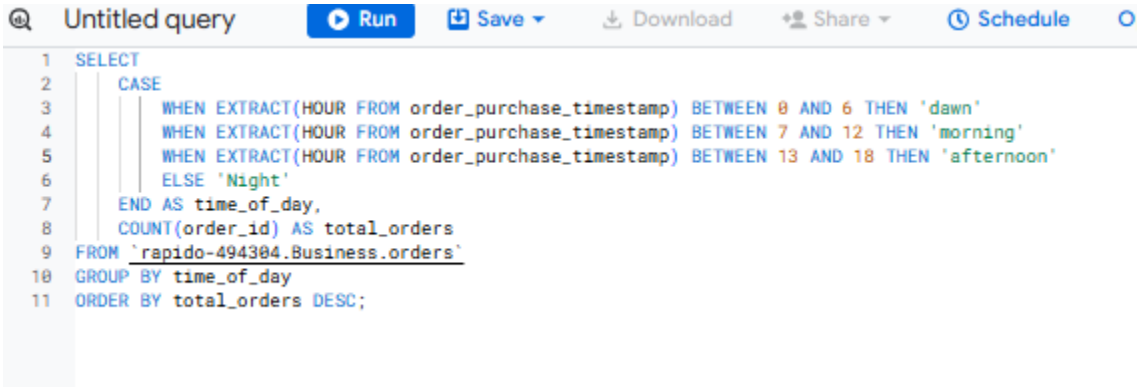
Business responses:-

- The number of orders changes from month to month. Some months have more orders, while others have less . This shows seasonal customer demands. The business can use this information to plan inventory, staffing, and marketing activities more effectively.

3.During what time of the day, do the Brazilian customers mostly place their orders? (Dawn, Morning, Afternoon or Night)

1. 0-6 hrs : Dawn
2. 7-12 hrs : Mornings
3. 13-18 hrs : Afternoon
4. 19-23 hrs : Night

SQL query: -



SQL responses:-



Row	time_of_day	total_orders
1	afternoon	38135
2	Night	28331
3	morning	27733
4	dawn	5242

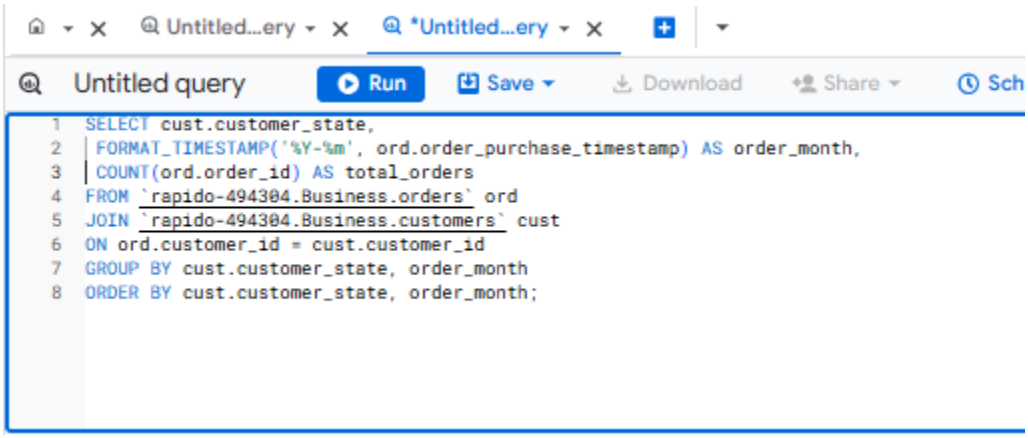
Business responses:-

- Most orders were placed during the Afternoon period. This helps the business identify peak customer activity hours and plan operations accordingly.

Evolution of E-commerce orders in the Brazil region :-

1.Get the month on month no. of orders placed in each state.

SQL query: -



SQL responses:-

Query results

Chat | Save results | Open i

Job information | Results | Visualisation | JSON | Execution details | Execution graph

Row	customer_state	order_month	total_orders
1	AC	2017-01	2
2	AC	2017-02	3
3	AC	2017-03	2
4	AC	2017-04	5
5	AC	2017-05	8
6	AC	2017-06	4
7	AC	2017-07	5
8	AC	2017-08	4
9	AC	2017-09	5
10	AC	2017-10	6
11	AC	2017-11	5

Business responses:-

- The report shows how order volume changed across different states over time. It helps identify states with high order activity, growth trends, and seasonal changes, supporting better business planning and forecasting.

2.How are the customers distributed across all the states?

SQL query: -

Untitled query Run Save Download

```sql
WITH customer_distribution AS (
    SELECT
        customer_state,
        customer_id
    FROM `rapido-494304.Business.customers`
)
SELECT
    customer_state,
    COUNT(DISTINCT customer_id) AS total_customers
FROM customer_distribution
GROUP BY customer_state
ORDER BY total_customers DESC;
```

SQL responses:-

Query results Chat Save resu

Job information | Results | Visualisation | JSON | Execution details | Execution graph

Row	customer_state	total_customers
1	SP	41746
2	RJ	12852
3	MG	11635
4	RS	5466
5	PR	5045
6	SC	3637
7	BA	3380
8	DF	2140
9	ES	2033
10	GO	2020
11	PE	1652

Results per page: 50 1 – 27 of

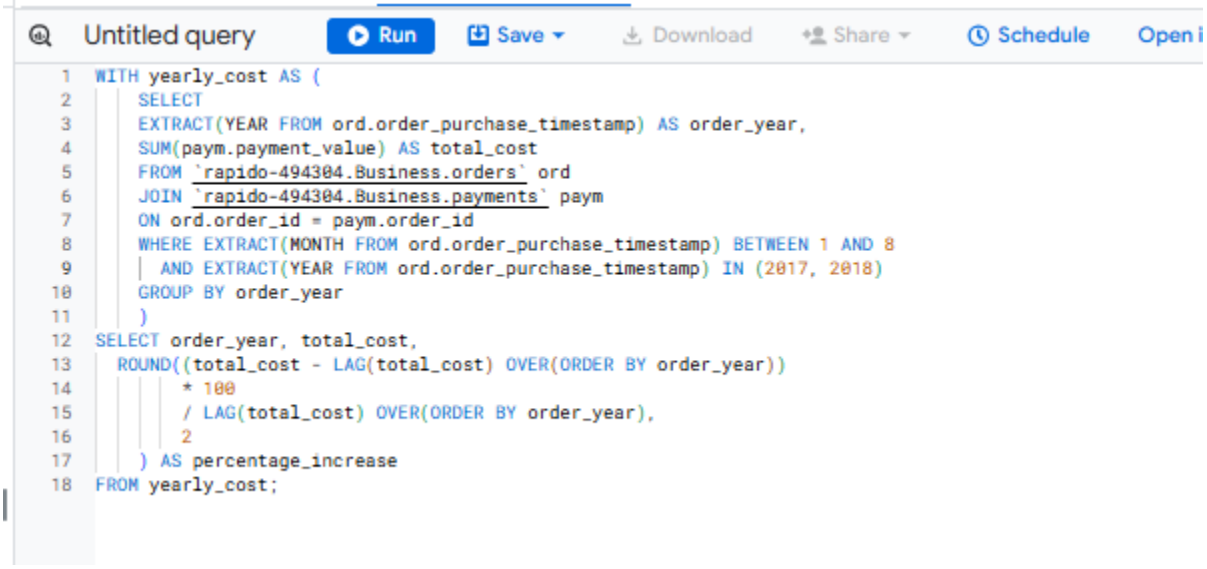
Business responses:-

- The report shows that SP has the highest number of customers, followed by RJ and MG. Most customers are concentrated in these states, making them important markets for business growth and planning.

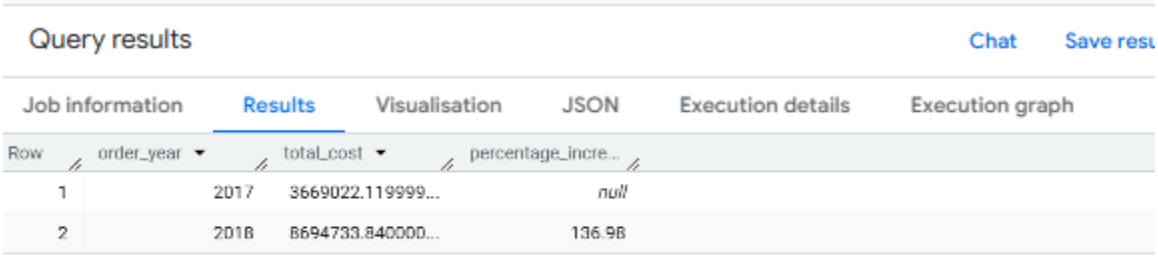
Impact on Economy: Analyze the money movement by e-commerce by looking at order prices, freight and others.

1.Get the % increase in the cost of orders from year 2017 to 2018 (include months between Jan to Aug only).
You can use the "payment_value" column in the payments table to get the cost of orders.

SQL query: -



SQL responses:-



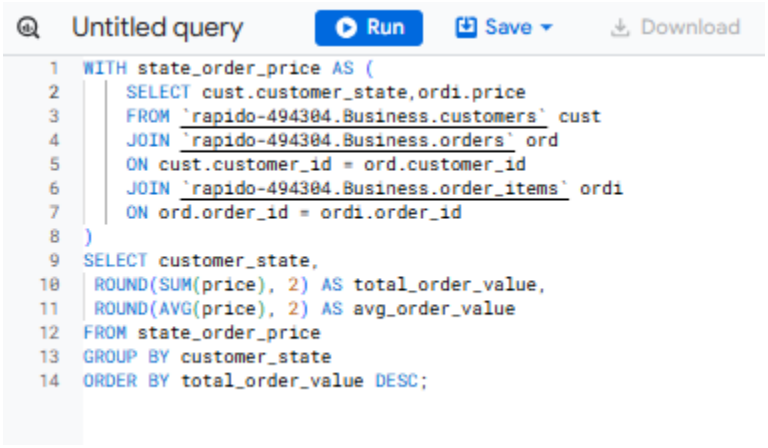
Row	order_year	total_cost	percentage_incre...
1	2017	3669022.119999...	null
2	2018	8694733.840000...	136.98

Business responses:-

- Percentage Increase = (Difference between 2018 cost and 2017 cost ÷ 2017 cost) × 100.
- The report compares total order value from January to August in 2017 and 2018. It shows how customer spending changed over the two years. A higher value in 2018 indicates business growth and increased customer purchases.

2.Calculate the Total & Average value of order price for each state.

SQL query: -




```sql
WITH state_order_price AS (
    SELECT cust.customer_state,ordi.price
    FROM `rapido-494304.Business.customers` cust
    JOIN `rapido-494304.Business.orders` ord
    ON cust.customer_id = ord.customer_id
    JOIN `rapido-494304.Business.order_items` ordi
    ON ord.order_id = ordi.order_id
)
SELECT customer_state,
 ROUND(SUM(price), 2) AS total_order_value,
 ROUND(AVG(price), 2) AS avg_order_value
FROM state_order_price
GROUP BY customer_state
ORDER BY total_order_value DESC;
```

SQL responses:-

Job information | Results | Visualisation | JSON | Execution details | Execution graph

Row	customer_state	total_order_value	avg_order_value
1	SP	5202955.05	109.65
2	RJ	1824092.67	125.12
3	MG	1585308.03	120.75
4	RS	750304.02	120.34
5	PR	683083.76	119.0
6	SC	520553.34	124.65
7	BA	511349.99	134.6
8	DF	302603.94	125.77
9	GO	294591.95	126.27
10	ES	275037.31	121.91

Business responses:-

- The report shows total revenue and average order value across different states. States with higher revenue contribute more to the business, while higher average order values indicate customers spend more per order. These insights help identify important markets for business growth.

3.Calculate the Total & Average value of order freight for each state.

SQL query: -

Untitled query | Run | Save | Download | Share

```
WITH state_freight AS (
    SELECT cust.customer_state,ordi.freight_value
    FROM `rapido-494304.Business.customers` cust
    JOIN `rapido-494304.Business.orders` ord
    ON cust.customer_id = ord.customer_id
    JOIN `rapido-494304.Business.order_items` ordi
    ON ord.order_id = ordi.order_id
)
SELECT customer_state,
  ROUND(SUM(freight_value), 2) AS total_freight_value,
  ROUND(AVG(freight_value), 2) AS avg_freight_value
FROM state_freight
GROUP BY customer_state
ORDER BY total_freight_value DESC;
```

SQL responses:-

Query results

Chat

Job information | Results | Visualisation | JSON | Execution details | Execution g

Row	customer_state	total_freight_value	avg_freight_value
1	SP	718723.07	15.15
2	RJ	305589.31	20.96
3	MG	270853.46	20.63
4	RS	135522.74	21.74
5	PR	117851.68	20.53
6	BA	100156.68	26.36
7	SC	89660.26	21.47
8	PE	59449.66	32.92
9	GO	53114.98	22.77
10	DF	50625.5	21.04
11	ES	49764.6	22.06

Business responses:-

- The report shows freight costs across different states. States with higher freight values have more orders, while higher average freight costs indicate more expensive deliveries. This helps identify shipping patterns and improve logistics planning.

Analysis based on sales, freight and delivery time

1.Find the no. of days taken to deliver each order from the order's purchase date as delivery time.
Also, calculate the difference (in days) between the estimated & actual delivery date of an order.
Do this in a single query.

You can calculate the delivery time and the difference between the estimated & actual delivery date using the given formula:

- **time_to_deliver** = order_delivered_customer_date - order_purchase_timestamp
- **diff_estimated_delivery** = order_delivered_customer_date - order_estimated_delivery_date

SQL query: -

Untitled query | Run | Save | Download | Share | Schedule | Open in

```sql
WITH delivery_metrics AS (
    SELECT order_id,order_purchase_timestamp,order_delivered_customer_date,order_estimated_delivery_date
    FROM `rapido-494304.Business.orders`
    WHERE order_delivered_customer_date IS NOT NULL
)
SELECT order_id,
 DATE_DIFF(
 DATE(order_delivered_customer_date),
 DATE(order_purchase_timestamp),
        DAY
    ) AS time_to_deliver,
 DATE_DIFF(
 DATE(order_delivered_customer_date),
DATE(order_estimated_delivery_date),
        DAY
    ) AS diff_estimated_delivery
FROM delivery_metrics;
```

SQL responses:-

Query results | Chat

Job information | Results | Visualisation | JSON | Execution details | Execution gra

Row	order_id	time_to_deliver	diff_estimated_d...
1	65d1e226dfaeb8cdc42f665422...	36	-17
2	2c45c33d2f9cb8ff8b1c86cc28...	31	-29
3	1950d777989f6a877539f53795...	30	12
4	bfbd0f9bdef84302105ad712db...	55	36
5	98974b076b01553d49ee64679...	44	-7
6	c4b41c36dd589e901f6879f25a...	36	-15
7	d2292ff2201e74c5db154d1b7a...	30	-21
8	95e01270fcbae986342340010...	31	-20
9	ed8c7b1b3eb256c7dce0c7423...	45	-6
10	5cc475c7c03290048eb2e742c...	69	18

Business responses:-

- The report analyzes delivery performance by comparing actual and estimated delivery dates. Early deliveries indicate efficient logistics, while delayed deliveries may affect customer satisfaction. These insights help improve delivery operations.

2.Find out the top 5 states with the highest & lowest average freight value.

SQL query: -

Untitled query Run Save Download Share

```sql
WITH state_freight AS (
    SELECT cust.customer_state,
    ROUND(AVG(ordi.freight_value), 2) AS avg_freight_value
    FROM `rapido-494304.Business.customers` cust
    JOIN `rapido-494304.Business.orders` ord
    ON cust.customer_id = ord.customer_id
    JOIN `rapido-494304.Business.order_items` ordi
    ON ord.order_id = ordi.order_id
    GROUP BY cust.customer_state
)
(SELECT ...
FROM state_freight
ORDER BY avg_freight_value DESC
LIMIT 5
)
UNION ALL
(
SELECT'Lowest' AS category, customer_state, avg_freight_value
FROM state_freight
ORDER BY avg_freight_value ASC
LIMIT 5
);
```

SQL responses:-

Job information | Results | Visualisation | JSON | Execution details | Execution graph

Row	category	customer_state	avg_freight_value
1	Highest	RR	42.98
2	Highest	PB	42.72
3	Highest	RO	41.07
4	Highest	AC	40.07
5	Highest	PI	39.15
6	Lowest	SP	15.15
7	Lowest	PR	20.53
8	Lowest	MG	20.63
9	Lowest	RJ	20.96
10	Lowest	DF	21.04

Business responses:-

- The report compares average freight costs across states. Higher costs indicate more expensive shipping, while lower costs suggest better logistics efficiency.

3.Find out the top 5 states with the highest & lowest average delivery time.

SQL query: -

```sql
WITH delivery_time AS (
    SELECT cust.customer_state,
     DATE_DIFF(
     DATE(ord.order_delivered_customer_date),
     DATE(ord.order_purchase_timestamp), DAY
       ) AS delivery_days
    FROM `rapido-494304.Business.orders` ord
    JOIN `rapido-494304.Business.customers` cust
    ON ord.customer_id = cust.customer_id
    WHERE ord.order_delivered_customer_date IS NOT NULL
),
avg_delivery AS (
    SELECT customer_state,
    ROUND(AVG(delivery_days), 2) AS avg_delivery_time
    FROM delivery_time
    GROUP BY customer_state
)
(
SELECT
    'Highest' AS category,
    customer_state,
    avg_delivery_time
FROM avg_delivery
ORDER BY avg_delivery_time DESC
LIMIT 5
)
UNION ALL
(
SELECT
    'Lowest' AS category,
    customer_state,
    avg_delivery_time
FROM avg_delivery
ORDER BY avg_delivery_time ASC
LIMIT 5
)
```

SQL responses:-

Query results

Chat | Save results | Open in

Job information | Results | Visualisation | JSON | Execution details | Execution graph

Row	category	customer_state	avg_delivery_time
1	Highest	RR	29.34
2	Highest	AP	27.18
3	Highest	AM	26.36
4	Highest	AL	24.5
5	Highest	PA	23.73
6	Lowest	SP	8.7
7	Lowest	PR	11.94
8	Lowest	MG	11.95
9	Lowest	DF	12.9
10	Lowest	SC	14.91

Business responses:-

- The report compares average delivery time across states. States with longer delivery times may face logistics challenges, while states with shorter delivery times show more efficient delivery operations. These insights help improve delivery performance and customer satisfaction.

4.Find out the top 5 states where the order delivery is really fast as compared to the estimated date of delivery.

You can use the difference between the averages of actual & estimated delivery date to figure out how fast the delivery was for each state.

SQL query: -

Untitled query | Run | Save | Download | Share | Schedule

```sql
WITH delivery_metrics AS (
    SELECT cust.customer_state,
    DATE_DIFF(DATE(ord.order_delivered_customer_date),DATE(ord.order_purchase_timestamp),DAY
        ) AS actual_delivery_days,
DATE_DIFF(DATE(ord.order_estimated_delivery_date),DATE(ord.order_purchase_timestamp),DAY
        ) AS estimated_delivery_days
FROM `rapido-494304.Business.orders` ord
JOIN `rapido-494304.Business.customers` cust
ON ord.customer_id = cust.customer_id
WHERE ord.order_delivered_customer_date IS NOT NULL
)
SELECT customer_state,
ROUND(AVG(actual_delivery_days), 2) AS avg_actual_delivery_days,
ROUND(AVG(estimated_delivery_days), 2) AS avg_estimated_delivery_days,
ROUND(AVG(actual_delivery_days) - AVG(estimated_delivery_days),2) AS delivery_speed
FROM delivery_metrics
GROUP BY customer_state
ORDER BY delivery_speed
LIMIT 5;
```

SQL responses:-

Query results | Chat | Save results | Open in

Job information | Results | Visualisation | JSON | Execution details | Execution graph

Row	customer_state	avg_actual_delive...	avg_estimated_d...	delivery_speed
1	AC	21.0	41.72	-20.72
2	RO	19.28	39.39	-20.1
3	AP	27.18	46.87	-19.69
4	AM	26.36	45.92	-19.57
5	RR	29.34	46.63	-17.29

Business responses:-

- The report identifies the states where orders are delivered earlier than the estimated date. Early deliveries indicate efficient logistics and strong delivery performance. This helps improve customer satisfaction and build customer trust.

Analysis based on the payments

1. Find the month on month no. of orders placed using different payment types.
   SQL query: -

```sql
WITH payment_orders AS (
    SELECT
     FORMAT_TIMESTAMP('%Y-%m', ord.order_purchase_timestamp) AS order_month, pay.payment_type, ord.order_id
    FROM `rapido-494304.Business.orders` ord
    JOIN `rapido-494304.Business.payments` pay
    ON ord.order_id = pay.order_id
)
SELECT order_month,payment_type,
  COUNT(DISTINCT order_id) AS total_orders
FROM payment_orders
GROUP BY order_month, payment_type
ORDER BY order_month, payment_type
```

   SQL responses:-

Query results

Chat | Save results | Op

Job information | Results | Visualisation | JSON | Execution details | Execution graph

Row	order_month	payment_type	total_orders
1	2016-09	credit_card	3
2	2016-10	UPI	63
3	2016-10	credit_card	253
4	2016-10	debit_card	2
5	2016-10	voucher	11
6	2016-12	credit_card	1
7	2017-01	UPI	197
8	2017-01	credit_card	582
9	2017-01	debit_card	9
10	2017-01	voucher	33

Business responses:-

- The report analyzes payment methods used by customers over time. It helps identify the most popular payment options and changes in customer payment preferences. These insights support better payment-related business decisions.

2.Find the no. of orders placed on the basis of the payment installments that have been paid.
   SQL query: -

Untitled query Run Save Download

```sql
WITH installment_orders AS (
    SELECT order_id, payment_installments
    FROM `rapido-494304.Business.payments`
)
SELECT payment_installments,
  COUNT(DISTINCT order_id) AS total_orders
FROM installment_orders
GROUP BY payment_installments
ORDER BY payment_installments
```

SQL responses:-

Query results

Job information | Results | Visualisation | JSON

Row	payment_installm...	total_orders
1	0	2
2	1	49060
3	2	12389
4	3	10443
5	4	7088
6	5	5234
7	6	3916
8	7	1623
9	8	4253
10	9	644

Business responses:-

- The report analyzes customer payment installment preferences. A higher number of orders with fewer installments indicates customers prefer quicker payments, while more installments suggest customers prefer flexible payment options. These insights help improve payment offerings.